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PI Name:	Hammond, Timothy M.D.		
Project Title:	Fuel to Mars		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-Artemis1 (EM1). App A: Orion (Artemis-1) (formerly Exploration Mission-1)
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/17/2024 per NSSC information (Ed., 1/4/24). NOTE: End date changed to 03/17/2023 per NSSC information (Ed., 9/22/22).		
Key Personnel Changes/Previous PI:	January 2021 report: None.		
COI Name (Institution):	Allen, Patricia M.S. (Veterans Health Administration) Birdsall, Holly M.D., Ph.D. (Institute For Medical Research, Inc.) Giaever, Guri Ph.D. (University of British Columbia, Canada) Nislow, Corey Ph.D. (University of British Columbia, Canada)		
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Performance Goal No.:	
Performance Goal Text:	NOTE: End date changed to 3/17/2022 per NSSC information (Ed., 5/18/21)
Task Description:	The primary research product this project will produce is identification of the insertion strains of Chlamydomonas reinhardtii, a single-cell green alga, which optimizes survival during exposure to the combined impacts of space radiation and microgravity in the cosmic space environment beyond the Van Allen Belts. The Chlamydomonas reinhardtii strains in a mapped insertional mutant library are a transformative tool allowing near genome wide analysis of the genes mediating survival advantage during stimuli such as cosmic radiation and microgravity. The secondary set of research products this project will produce are identification of the pathways mediating the gene changes, visualization of the pathway interactions, and identification of any environments giving similar gene pathway signatures in other model systems. Our goal is to identify strains of Chlamydomonas reinhardtii with gene insertions, which convey the best survival advantage during the exposure to the combined impacts of space radiation and microgravity for future use as parent strains in hydrogen production initiatives. Hydrogen is the lead contender for fuel production for return space flight missions from orbital bodies beyond low Earth orbit. The metabolism of select microorganisms makes hydrogen fuel. The best-known and most efficient hydrogen producer is Chlamydomonas reinhardtii, which is poised for exploitation as an industrial biotechnology platform for biofuel production. The type of investigation we propose is experiments that will be conducted aboard the Exploration Mission-1 (EM-1) [Note now called Artemis] lunar orbit flight, with asynchronous controls in Kennedy Space Center ISSES (International Space Station Environmental Simulators) Chambers. The proposed hardware is the NASA Biological Research in Canisters (BRIC)-100VC. Hypothesis: The hypothesis to be tested is that the mapped insertional mutant library of Chlamydomonas reinhardtii will allow identification of the genes that mediate optimal radiation resistance, and imperviousness to microgravity, in the cosmic space environment beyond the Van Allen Belts. The specific aims of the proposal are: Specific Aim 1: To experimentally determine which of the thousands of mapped insertion mutant strains of Chlamydomonas reinhardtii have the best long-term survival advantage during combined space radiation and microgravity impacts in the cosmic space environment beyond the Van Allen Belts. Specific Aim 2: To analyze the dataset of insertion strains with the best long-term survival advantage during combined space radiation and microgravity impacts to: • Perform pathway analysis using gene set enrichment analysis (GSEA); • Visualize the pathway interactions using Cytoscape/ClueGo; • Make comparison to homologous gene sets in Chlamydomonas, Saccharomyces cerevisiae, and other model organism databases for any environments giving similar gene pathway signatures. Although focused on gene pathways for survival for future applications in hydrogen fuel production, the data generated will provide genome level insights into the cancer risks of prolonged exposure to cosmic radiation and microgravity. The spectrum of cancer risk genes is known, and this study will parse the importance of risk genes, and the patterns of radiation risk change with exposure to the cosmic space environment beyond the Van Allen Belts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research impacts and Earth benefits include: o The first opportunity to apply modern “omics” techniques to samples flown beyond the van Allen belts, and recover for ground-based analysis. o The first biological validation of the utility of the Chlamydomonas deletion series for academic and commercial applications. o Delineation of the production of the potential fuels, lipid and hydrogen, to optimize use in future space based and ground applications. o Education of the next generation of space flight investigators. o Identifies mechanisms of radiation damage due to cosmic radiation with possible future clinical applications. o This proposal directly addresses the Space Biology Plan Element GL3: What are the key cellular and molecular systems necessary for biological organisms to thrive in space environment that can be discovered from 21st century bioinformatics tools? o This application directly addresses three elements of the Human Research Roadmap on Risk of Radiation Carcinogenesis: Cancer 07: How can systems biology approaches be used to integrate research on the molecular, cellular, and tissue mechanisms of radiation damage to improve the prediction of the risk of cancer and to evaluate the effectiveness of countermeasures? How can epidemiology data and scaling factors support this approach? Cancer 08: What are the most effective medical or dietary countermeasures to mitigate cancer risks from exposure to SPE (solar particle events) and GCR (galactic cosmic rays)? What side effects should be tolerated versus mission risks? Can medical countermeasures that provide mitigation across multiple cancer types or across multiple radiation risk areas be identified? Cancer 09: Are there significant effects from other spaceflight factors that modify the carcinogenic risk from space radiation?

OVERALL PROGRESS DURING THE PERFORMANCE PERIOD
Chlamydomonas reinhardtii preparation for experiments.

Chlamydomonas reinhardtii (CC-125 wild type mt+ [137c]) and the second generation mapped insertional deletion series were purchased from the University of Minnesota Chlamydomonas collection (Minneapolis, MN). The first generation mapped insertional mutant library was a kind gift of Dr. Martin Jonikas (Princeton University). The Chlamydomonas mapped insertional mutant library of 13,000 strains was received, spotted on Tris-acetate-phosphate (TAP) agar plates, and aliquots re-seeded onto fresh plates robotically every 2 weeks. TAP was 20 ml 1M Tris base, 1ml phosphate K2HPO4/KH2PO4 buffer, 1ml Hunter's trace metals, 10ml of solution A [NH4Cl, MgSP4.7H2O, 1ml CaCl2.2H2O], glacial acetic acid pH 7 made up to 1 liter with distilled water.

Chlamydomonas, with an insertion of the gene encoding Dsup, were a gift of C. Nislow (University of British Columbia). Dsup is a tardigrade damage suppressor protein known to protect animal cells against gamma and ionizing radiation in ground-based studies. Dsup gene location to the nucleus was confirmed by including green fluorescent protein (GFP) in the plasmid, and localization by fluorescent microscopy.

Flight hardware design and build.

Moonshot hardware was designed and built by The Bionetics Corporation (NASA Kennedy Space Center, Brevard County, FL). Bionetics custom "Moonshot" hardware is an ultralow-power growth system for use on Artemis-1 (Hammond et al 2021). Each of the three growth modules holds one standard 100 mm diameter circular Petri dish. The plates are illuminated with blue (~450 nm) and red (~660 nm) light-emitting diode (LED) lights continuously for 6 h in every 24 h. The combination of red and blue light is optimum for Chlamydomonas. The hardware monitors and records the temperature of the experiment, the power level, and draw from the batteries (as an indication that the lights were turned on and off), as well as the acceleration levels in three axes. Once activated and loaded with the biological samples, the flight hardware was placed in a custom lathed form-fitting Styrofoam support, and encased in a Nylex bag.

Radiation measurements.

Crew Active Dosimeter (CAD) and Radiation Area Monitor (RAM) radiation detectors were placed adjacent to the biological flight hardware. The detectors had been assembled and delivered to NASA Kennedy Space Center prior to flight by the Space Radiation Analysis Group based at NASA Johnson Space Center (JSC) by Human Health and Performance Directorate/Leidos personnel. The detectors were collected post-flight and data was analyzed at NASA JSC.

Artemis-1 flight.

After an initial delay of 19 days on the launch pad, the Artemis-1 mission was launched from NASA Kennedy Space Center (KSC) on November 16, 2022 for a planned 25-day space mission. Orion completed one flyby of the Moon on November 21, followed by a distant retrograde orbit for six days and then a second flyby of the Moon on November 25, and subsequently returned to Earth. The Orion capsule was recovered from the Pacific Ocean, returned to California, and transported back to KSC by truck. However, the Space Biology experiments were first removed from Orion in California, flown to Kennedy Space Center (without security radiation scanning) and released to the investigators nine days after the landing. The samples for the payload described in the report were in the hardware for a total of 56.5 days.

ANALYSIS

Spectrophotometry.

Spectrophotometry was performed on a Molecular Dynamics (San Jose CA) M5e Spectrophotometer. Chlamydomonas colonies were scraped from the agar, washed three times in phosphate buffered saline, re-suspended in TAP media, and assayed by spectrophotometry for Chlorophyll A (absorbance at 436 nm), Chlorophyll B (absorbance at 472 nm), Photo system I and II (absorbance at 700 nm), Cytochrome f (absorbance at 554 nm), and protein content (absorption at 800 nm). Net absorbance for each wavelength was calculated by subtracting the background absorbance of TAP buffer alone. The net absorbance values were then normalized to the protein content as measured by absorbance at 800 nm.

Flow cytometry.

Flow cytometry was performed on a Becton-Dickinson Accuri C+ Plus flow cytometer (Franklin Lakes, NJ). Six colonies from flight and ground plates, plus four D-sup colonies from flight and ground plates, were harvested and re-suspended in annexin-binding buffer, which is a phosphate buffer with calcium. The aggregates were disrupted by vortexing and then filtered through a 70µm nylon mesh to remove clumps that would plug the flow cytometer. The resuspended Chlamydomonas samples were aliquoted 100 µl per well into V-bottom microtiter plates. Individual aliquots were either stained with Alexa Fluor 488-annexin (5 µl/well) and propidium iodide (PI) (1 µl) to identify apoptosis and necrosis, or Nile Red (1:100 dilution from a stock solution of 100 µg ml-1 in methanol) with 0.005% Triton-X (1:100 dilution from a stock solution of 0.5% in water) permeabilization to measure lipid content. We counted 2,000 Chlamydomonas cells from each sample. Background signal was estimated by appropriate no-dye controls. In every flow cytometry run, quality controls were performed on the instrument with 3 color fluorescent beads, followed by assay of 5 control tubes: (1) no dyes, (2) Alexa fluor 488 annexin binding alone, (3) propidium iodide alone, (4) Alexa fluor 488 annexin binding with propidium iodide, and (5) Nile red alone. Flow cytometric and spectrophotometric data were evaluated by two-tailed Student's t-test comparing heat- or cold-exposed Chlamydomonas to the relevant room temperature control.

Single-cell confocal Raman spectroscopy of Chlamydomonas reinhardtii.

Multiple colonies of Chlamydomonas were harvested from a ground plate and flight plate. The samples included colonies flown live and colonies pretreated with azide before flight. The colonies were harvested into 1.5ml Eppendorf tubes with 500 µL TAP, and re-suspended by pipetting and vortexing. The resultant Chlamydomonas solution was filtered through a 70 µm mesh into 12 x 74 mm polypropylene tubes to remove multicellular aggregates. The remaining solution of cells was divided into two 500µL Eppendorf tubes, spun at 3000g for 1 minute to pellet the cells, and the supernatant aspirated. One tube was left to air dry. The second tube had 100 µL of 10% electron microscopy paraformaldehyde added, incubated for 10 min, and then was spun to pellet the cells before the supernatant was aspirated. This paraformaldehyde fixed sample was washed with 500 µL deionized water, spun, and the supernatant aspirated.

Task Progress:

A 2 µl aliquot of each sample rehydrated with distilled water was spotted on quartz slide (25 mm x 25 mm x 1 mm) that was kept on a microscope glass slide covered with aluminum foil. This substrate showed minimal background signal compared to various other substrates we tried such as glass, CaF2, and Al2O3. A silicon wafer was not used, as it has strong Raman band at 521 cm-1 and interfered with the current analysis. The samples were air dried at room temperature and were analyzed. A Horiba LabRAM Odyssey Raman microscope was used for the Raman spectral characterization of these samples. Laser power density was optimized in order to achieve better spectral intensity with characteristic Raman bands, which was not present at lower laser power densities. Exposure time and laser intensity was optimized by conducting a series of experiments to prevent charring of the samples, thereby to eliminate any unwanted signals due to laser-induced biomolecule denaturation. A 785 nm laser was used to acquire Raman spectra (acquisition time of 30 s with 300 lines/mm grating). First, white light focused on the individual algal cells at a magnification of 100X. Then, Raman spectra were collected in the range of 400-1800 cm-1.The spectral baselines were pre-processed by polynomial fitting.

STATISTICS

Flow cytometric and spectrophotometric data are presented as geometric mean ± standard error with six replicates (unless otherwise noted). Correlations were analyzed by Statistica 6.1 (StatSoft Inc., Tulsa, OK) using correlation matrix product moment and partial correlations. For the Raman spectral analysis: Spectra were collected from five or more algal samples across three different colonies exposed to various conditions and subsequently averaged to better represent statistical variations in the intensities of vibrations corresponding to various biomolecular components within the spectra. The standard deviation of these intensities under different conditions was illustrated using error bars in the histogram to provide a clear visualization of the variability and reliability of our measurements.

RESULTS

Spectrophotometry of Chlorophyll a and b, Cytochrome f, and Photosystem levels.

Levels of the light absorbing pigments chlorophyll A and B, the mitochondrial heme protein cytochrome c, and the photon harvesting pigment photosystems were no different between ground and flight samples.

Radiation exposure in assorted environments in space and on Earth.

The total radiation exposure during the flight as measured in the RAM and CAD were compared to the exposure levels during flight to other environmental systems. Radiation Area Monitor Mission Dose (mGy) 11.4 ± 0.4; Crew Active Dosimeter Mission Dose (mGy) 12.6 ± 0.5; Naturally occurring "background" radiation (Earth) (mGy) 0.1; CAT-Scan of Abdomen & Pelvis, + contrast (mGy) 15.4; Exposure on Mars for 25.5 days (Curiosity Rover) (mGy) 5.4; Mars transit for 25.5 days (Curiosity Rover) (mGy) 11.8; and ISS for 25.5 days in (mGy) 5.7.

Flow cytometry assay of viability and apoptosis of Chlamydomonas in Flight vs Ground.

The percentages of apoptotic cells (Annexin V positive) and dead cells (propidium iodide positive) in the flight and ground control samples showed significantly ($p<0.0001$) increased apoptosis and decreased cell death in the flight samples.

Flow cytometry assay of the effect of Dsup on the viability of Chlamydomonas in Flight vs Ground. Aliquots of Chlamydomonas had the tardigrade D-Sup gene, known to be protective during diverse ground-based radiation exposures, transfected into their nuclei. A GFP marker was included in the D-sup cassette. Nuclear D-sup localization and expression was confirmed by fluorescent microscopy. The D-Sup transfected Chlamydomonas had better survival both in flight and on the ground compared to irrelevant control transfected strains. The percentages of live cells (PI negative) in flown and ground control Chlamydomonas transfected with nuclear D-sup showed increased cell survival in both group and flight sample sets.

Single-cell confocal Raman spectroscopy of Chlamydomonas.

In a pioneering use of Raman spectroscopy for the analysis of space biology samples, we noticed a differential survival strategy among Chlamydomonas exposed to cosmic rays and/or microgravity. Specifically, we found two different cell types (Type I and Type II) with distinct Raman spectral features in flight samples compared to the ground samples. The intensity of Raman vibration corresponding to carotenoids in Type I cells (FL-Type I) was much weaker; whereas, in Type II cells (FL-Type II), characteristic carotenoid vibrations enhanced significantly. In Raman spectra of Chlamydomonas, the presence of prominent vibrations at 1001, 1156, and 1523 cm-1 indicate the presence of carotenoids, which are attributed to C–CH3 deformation, C-C stretching, and C=C stretching vibrations of polyene, respectively. The significant variation in these carotenoid Raman vibrations found in the spectral profiles between Type I and Type II algal cells offers a compelling insight into the heterogeneity of stress responses within a single-cell population in two-cell populations. The lack of significant carotenoid vibrations in the Type I flight sample implies that these cells may not have been able to induce the same protective response and might have yielded to radiation-induced damage. Moreover, the azide-killed flight samples exhibited nearly identical Raman spectra with weak carotenoid signal as in Type I cells, corroborating our proposed link between the absence of carotenoid and cell death pathways. The weaker carotenoid signals in these samples suggest that the living cells in the flight environment actively maintain or increase their carotenoid content. In contrast, the azide treatment or cell death ceases metabolic activity, thereby reducing the carotenoid levels.

For better understanding the relative increase in the carotenoid content at various conditions, the intensity ratio (ICaro/Chlo) of carotenoid-chlorophyll vibration (1523 cm-1 and 1359 cm-1, respectively) was used. We also used Raman spectroscopy to identify any significant changes in chlorophyll composition by analyzing prominent and characteristic Raman bands of chlorophylls found at 1359 (vibrations of C-C and C-N bonds within the porphyrin ring) and 1549 cm-1 (primarily involves the C=C stretching vibrations within the porphyrin ring). We noticed that the intensity ratio of these bands (I1549/1359) in flight samples (FL-Type I and FL-Type II), as well as in azide-killed samples, showed no significant differences compared to the ground sample, suggesting comparable chlorophyll levels across these conditions. This was in concordance with the data obtained from the spectrophotometry. A slight variation in the I1549/1359 found in FL-Type II could potentially be due to the variation in Chlorophyll a to Chlorophyll b ratio, as vibration corresponds to the stretching vibrations of the carbonyl (C=O) group in the aldehyde group present in Chlorophyll b also contributes to the intensity of Raman vibration at 1549 cm-1 (this ratio has slightly increased in FL-Type II sample). In addition, a minor shift in the 1549 cm-1 band to 1546 cm-1 in FL-Type II samples was also found, which could possibly be attributed to the change in the electronic environment around the chlorophyll molecules

	as an adaptive response to radiation stress and subsequent variation in Chlorophyll b content. These spectra and ratios provide insights into the biochemical adaptations of Chlamydomonas to galactic radiation and microgravity, with significant variations noted between the two cell types we found in live flight samples, suggesting differential stress responses and survival strategies. Moonshot (new flight hardware). The Moonshot hardware performed nominally throughout the flight and ground control experiments. The maximum temperature delta between flight and ground control hardware copies was 2.5 oF, well within the envelope of tolerable experimental conditions. At the end of the 25.5-days flight, the batteries still had more than 50% of power remaining. The power profile validated that the lights came on for 6 out of 24 h daily, as planned. Acceleration recordings show serial measurements of gravity in the z axis, but no acceleration in other directions as expected. The hardware met all the validation criteria for power use, power cycling, temperature control, g-force recording in three axes, data capture, and downloading for analysis, materials, and biological compatibility. The Moonshot hardware is now not only flight certified, but flight proven.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	Khoshnoodi M, Barazandeh M, Porter I, Allen P, Birdsall H, Hammond T, Nislow C. "Improved drug toxicity testing: hepatocyte spheroids in 3D Cell Spinpod rotating suspension culture." 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. Abstracts. 38th Annual Meeting of the American Society for Gravitational and Space Research, Houston, TX, November 9-12, 2022. , Nov-2022
Articles in Peer-reviewed Journals	Hammond TG, Panikkanvalappil SR, Allen PL, Gaikani HK, Nislow C, Giaever G, Zhang Y, Levine HG, Gaza R, Dimapilis D, Wells HW, Russick JM, Durand PM, Birdsall HH. "Programmed cell death and redox metabolism protect Chlamydomonas reinhardtii populations from the galactic cosmic environment on the Artemis-1 mission." Sci Rep. 2025 Jul 2;15:23396. https://doi.org/10.1038/s41598-025-05419-w ; PMID: 40603369; PMCID: PMC12222459 , Jul-2025